Invigilator's Signature and Date

Set-1

KIIT DEEMED TO BE UNIVERSITY

Autumn End Semester Examination-2022

Roll No.	
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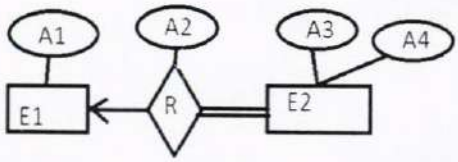
RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-A

(Answer All Questions)

Time: 30 Minutes **Full Marks = 2 × 7 = 14 Marks**

Question No	Question	Write the correct option here.
Q.No:1	Explain integrity in context of database system. **Answer:**	
Q.No:2	Explain physical data independence. **Answer:**	
Q.No:3	 What is the minimum number of tables required to represent this situation in the relational model? A. 2 B. 3 C. 4 D. 5	

Q.No:4	Explain multi-valued attributes with an example. **Answer:**	
Q.No:5	Explain the use of aggregate functions in SQL. **Answer:**	
Q.No:6	What happens in natural join operation? **Answer:**	
Q.No:7	What is partial dependency? Explain with example. **Answer:**	

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RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-B

(Answer Any Three Questions.)

Time: 1 Hour and 30 Minutes **Full Marks = 12 × 3 = 36 Marks**

Q.No:8 Consider the following tables and give both relational algebra expression and relational calculus expression of the queries.

Employee (emp_id, f_name, l_name, address, dept_no)

Department (dept_no, dept_name, manager_id)

Project (proj_id, proj_location, dept_no)

Works_on (emp_id, proj_id, hours)

Dependent (emp_id, dependent_name, age, relation)

a. Retrieve the name and address of all employees who work for the research department. (2+2)

b. For every project located at 'Delhi', list the project number and the controlling department number, and the department manager's last name and address. (2+2)

c. Find out the managers who have no dependents. (2+2)

Q.No:9 Consider the following relation for published books:

Book (Book_title, Author_Name, Book_Type, listprice, author_affil, Publisher)

Author_affil refers to the affiliation of author.

Suppose the following FDs exist :

Book_title → Publisher, Book_type Book_Type → Listprice

Author_Name → Author_affil

(a) What normal form is in the relation ? Explain your answer.

(b) Normalize until you cannot decompose the relation further using FD.

(c) Indicate the PK&FK. (3 + 6 +3)

Q.No:10 Explain 3-schema architecture of DBMS and it's advantages. [5]

Draw a simple ERD of an educational institute (Ex: KIIT-DU) with entities like student, faculty, department etc. Then, convert that ERD into schema diagram. [7]

Q.No:11 1) Concurrency control helps in managing simultaneous operations without conflicting with each another. In relation to this, discuss with example the following problems that may be encountered during concurrency execution of transactions:

I. Lost update problem

II. Dirty read problem. (6 marks)

2) Check whether the given schedules are conflict serializable or not using precedence graph.

a) R4(X), R2(X), R3(X), W1(Y), W2(X), R3(Y), W2(Y)

b) R1(X), R2(Z), R1(Z), R3(X), R3(Y), W1(X), W3(Y), R2(Y), W2(Z), W2(Y)

(6 marks)

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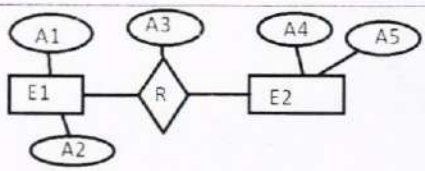
RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-A

(Answer All Questions)

Time: 30 Minutes **Full Marks = 2 × 7 = 14 Marks**

Question No	Question	Write the correct option here.
Q.No:1	Explain atomicity in context of database system. **Answer:**	
Q.No:2	Explain logical data independence. **Answer:**	
Q.No:3	 What is the minimum number of tables required to represent this situation in the relational model? (A) 2 (B) 3 (C) 4 (D) 5	

Q.No:4	Explain complex attributes with an example. **Answer:**	
Q.No:5	Explain the use of views in SQL. **Answer:**	
Q.No:6	What happens in left-outer join operation? **Answer:**	
Q.No:7	What is transitive dependency? Explain with example. **Answer:**	

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Autumn End Semester Examination-2022

RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-B

(Answer Any Three Questions.)

Time: 1 Hour and 30 Minutes **Full Marks = 12 × 3 = 36 Marks**

Q.No:8 Consider the following relations :
HOTEL (hotelno, name, address)
ROOM (roomno, hotelno, type, price_pn)
BOOKING (hotelno, guestno, dateform, dateto, roomno)
GUEST (guestno, name, address)
where the underlined column names are primary keys.

Write down expressions in both relational algebra and calculus for the following queries:

a. List all the hotels which are situated in Kolkata. (2+2)

b. List all single rooms with a charge below Rs. 1000 pernight. (2+2)

c. List all guests currently staying at Taj Hotel. (2+2)

Q.No:9 (a) Why do we normalize a relation ? How many different type of normal forms are there?

(b) Each student has unique roll. In student database for each student following information is to be stored : Roll, Name, Address, Dept. Code, Dept. Name and also for each subject in which he / she appears for examination store sub code, sub name, full marks, pass marks, score, further assume the following functional dependencies :

Roll → Name , address, dept code
Dept code →dept name
Sub code → sub name, full marks, pass marks
Roll, sub code → score.

Normalize the database up to 3NF showing the steps. Indicate primary and foreign keys.

(3 + 9)

Q.No:10 With help of a neat block diagram explain the basic components of a DBMS. [5]

Draw a simple ERD of a banking system with entities like branch, customer, employee, loan etc. Then, convert that ERD into schema diagram. [7]

Q.No:11 1. Explain how serializability help to maintain the consistency of the database. Verify whether the given schedule S is conflict serializable or not, using precedence graph.
S : R1(A) , R2(A) , R1(B) , R2(B) , R3(B) , W1(A) , W2(B) (6 marks)

2.

a) Using label precedence graph, check whether the given schedule S with transactions T1, T2, and T3 is view serializable or not.

S= R1(A), R2(A), W3(A), W1(A) (3 marks)

b) Give your comment on the state of the following schedule with justification. (3 marks)

T1	T2
Lock-X(A)	
R(A)	
W(A)	
	Lock-X(B)
	R(B)
	W(B)
	Lock-X(A)
Lock-X(B)	

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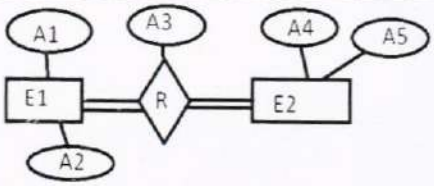
RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-A

(Answer All Questions)

Time: 30 Minutes **Full Marks = 2 × 7 = 14 Marks**

Question No	Question	Write the correct option here.
Q.No:1	What do you mean by inconsistent database? **Answer:**	
Q.No:2	Explain the importance of data models. **Answer:**	
Q.No:3	 What is the minimum number of tables required to represent this situation in the relational model? A. 3 B. 1 C. 2 D. 4	

Q.No:4	Explain cardinality with a proper example. **Answer:**	
Q.No:5	What is the difference between WHERE and HAVING Clause? **Answer:**	
Q.No:6	What happens in right-outer join operation? **Answer:**	
Q.No:7	What do you mean by Normalization? Explain with example. **Answer:**	

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Autumn End Semester Examination-2022

RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-B

(Answer Any Three Questions.)

Time: 1 Hour and 30 Minutes **Full Marks = 12 × 3 = 36 Marks**

Q.No:8 Consider the following tables:

Match(Matchid, Team1, Team2, ground, date, winner)

Player(Playerid, Lname, Fname, Country, YBorn, Bplace, Ftest)

Batting(Matchid, Pid, Over, runs, balls, Four, Sixes)

Bowling(Matchid,Pid,Overs,Maidens, Runs, Wickets)

Write down expressions in both relational algebra and calculus for the following queries:

A. List the name of players who belongs to'Australia'. (2+2)

B. Find the information of players who has born after 1980. (2+2)

C. Display a sorted list of ground names where England has played as team1. (2+2)

Q.No:9 (a) Why is normalization necessary? Compare between BCNF and 3rd Normalform. (4)

(b) Consider the relation R {RN, POR, PI, PN, DATE, MA, MT} and Functional Dependencies: { RN → POR; PI → PN, RN; PI, DATE → MA, MT, POR }

To which normal form does this belong? Decompose the relation so that it can belong to 3NF. Also show that the decomposition is lossless and it preserves dependency. (8)

Q.No:10 List the advantages of DBMS over traditional file processing systems. (5)

Draw a simple ERD of a library management system with entities like members, librarians, books etc. Then, convert that ERD into schema diagram. (7)

Q.No:11 1. Check the serilaizability of the following schedules using precedence graph/ label precedence graph as per the question. (6)

a) R2(X), W3(X), W1(X), W2(Y), R4(X), R4(Y), for conflict serializability.
b) R1(X); W1(X); R3(X); R2(X); W3(X), for view serializability.

2. Why concurrency control is needed? Explain lost update and dirty read problems with examples. (2+4=6 Marks)

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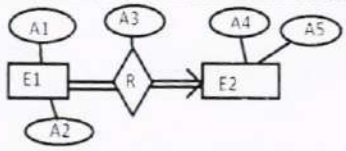
RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-A

(Answer All Questions)

Time: 30 Minutes **Full Marks = 2 × 7 = 14 Marks**

Question No	Question	Write the correct option here.
Q.No:1	Explain data redundancy in context of database system. **Answer:**	
Q.No:2	Explain the term relational in RDBMS. **Answer:**	
Q.No:3	 What is the minimum number of tables required to represent this situation in the relational model? A. 1 B. 2 C. 3 D. 4	

Q.No:4	Explain the importance of foreign key. **Answer:**	
Q.No:5	Explain the difference between UNION and UNION ALL with example. **Answer:**	
Q.No:6	What happens in full-outer join operation? **Answer:**	
Q.No:7	What is Pseudo-transitive dependency? Explain with example. **Answer:**	

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KIIT DEEMED TO BE UNIVERSITY

Autumn End Semester Examination-2022

RELATIONAL DATABASE MANAGEMENT SYSTEM (CS3044)

7th Semester B.Tech

SECTION-B

(Answer Any Three Questions.)

Time: 1 Hour and 30 Minutes **Full Marks = 12 × 3 = 36 Marks**

Q.No:8 Consider the following relations :
HOTEL (hotelno, name, address)
ROOM (roomno, hotelno, type, price_pn)
BOOKING (hotelno, guestno, dateform, dateto, roomno)
GUEST (guestno, name, address)
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Roll → Name , address, dept code
Dept code →dept name
Sub code → sub name, full marks, pass marks
Roll, sub code → score.

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S : R1(A) , R2(A) , R1(B) , R2(B) , R3(B) , W1(A) , W2(B) (6 marks)

2.

a) Using label precedence graph, check whether the given schedule S with transactions T1, T2, and T3 is view serializable or not.

S= R1(A), R2(A), W3(A), W1(A) (3 marks)

b) Give your comment on the state of the following schedule with justification. (3 marks)

T1	T2
Lock-X(A)	
R(A)	
W(A)	
	Lock-X(B)
	R(B)
	W(B)
	Lock-X(A)
Lock-X(B)	

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